K.R. MANGALAM UNIVERSITY

School of Engineering & Technology

SEC042 — NEW AGE PROGRAMMING LANGUAGES LAB

PHASE II
PROJECT IMPLEMENTATION & PROGRESS REPORT

PERSONAL FINANCE TRACKER

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Phase II documents the implementation progress, completed work, ongoing modules, and remaining tasks.

1. Phase II Overview

Phase II presents the current implementation of the Personal Finance Tracker Android application, shown in the attached screenshots under the app interface name “SachVault.” The visible interface is organized into Records, Analysis, Budgets, Accounts, and Categories areas. These screens are designed to let a user review transaction records, view income and expense summaries, manage budget information, view account information, and organize financial categories from one mobile application.

The current screenshots show that the main navigation and screen layouts for Records, Analysis, Budgets, Accounts, and Categories have been developed. The Analysis area is nearly complete. Budget, Accounts, and Categories are reported at approximately 80% completion. The next stage is to finish the remaining operations, verify that information is shared correctly between related screens, and test the application using actual transaction data.

Progress percentages are approximate development estimates. The screenshots show the current UI state; functional completion must be confirmed through testing.

2. Technology Stack

Technology	Role in the project
Kotlin	Primary language for application logic and event handling.
Android Studio	Development environment used to build and run the Android application.
XML	Native Android layouts and visual components.
Room Database / SQLite	Local persistence for application data.
RecyclerView	Presentation of lists such as transactions or categories, where used.
Material Components	Reusable Android interface components, where used.

List only technologies and libraries actually present in the project. Remove or relabel any component that has not been used.

3. Development Progress Summary

Module	Reported progress	Current status
Analysis	Nearly complete	Core analysis functionality is substantially implemented; final checks and refinements remain.
Budget	Approximately 80%	Budget screen and summary layout are present; complete remaining budget actions and verify calculations.
Accounts	Approximately 80%	Search and account-type filters are visible; finish account operations and verify saved account data.
Categories	Approximately 80%	All, Expenses, and Income filters are visible; finish category actions and verify category use.

3.1 Progress Snapshot

Analysis: Nearly complete

Budget: 80%

Accounts: 80%

Categories: 80%

Analysis is listed as nearly complete. Budget, Accounts, and Categories are each listed at approximately 80% completion.

4. Module-wise Implementation Status

4.1 Analysis Module — Nearly Complete

The Analysis screen provides a summary view of financial activity. In the supplied screenshot, it displays separate summary cards for Income, Expense, and Balance, together with navigation options for Income and Accounts and an Expense Breakdown area. The screenshot currently shows zero values and an empty expense breakdown, indicating that no sample transactions are visible in the captured state. The analysis calculations should therefore be verified after adding test transactions.

- The Analysis tab is visible in the application's bottom navigation and opens a dedicated analysis screen.
- The screen layout presents Income, Expense, and Balance summaries, with an area intended to show expense breakdown information.
- The summary values and breakdown should be calculated from the records stored in the application.
- Remaining checks include confirming the totals after adding, editing, or deleting records, and confirming that the empty state is displayed correctly when no transactions exist.

4.2 Budget Module — Approximately 80% Complete

The Budgets screen shows a month selector, summary cards for Total Budget and Total Spent, and an add button for starting a budget-related action. In the supplied screenshot, both displayed amounts are ₹0.00. This confirms that the budget screen and its main summary layout are present; the remaining implementation work is to complete the budget-entry flow and verify that budget totals and spending values update correctly when data is entered.

- The Budgets tab is visible in the application's navigation and opens the budget summary screen.
- The screen currently displays the selected month, Total Budget, Total Spent, and an add control. The full add/edit workflow should be confirmed through functional testing.
- The intended calculation is to compare the budget amount with the expenses recorded for the selected period.
- Remaining tasks include completing the budget creation and management actions, checking monthly calculations, and confirming that the summary refreshes after transaction changes.

4.3 Accounts Module — Approximately 80% Complete

The Accounts screen includes a search field, account-type filters labelled All, Bank, Card, and Cash, and an account summary area. The captured screen also shows an All Accounts amount and summary indicators, currently displaying zero values. The module is reported as approximately 80% complete. Remaining work is to finish the account-management interactions and verify that account information is stored, retrieved, and reflected correctly in the relevant parts of the application.

- Complete and verify the account actions provided by the interface, including adding or editing account details where those actions are supported.
- Check that account details remain available after navigating away from the screen or restarting the application.
- Verify that the account-type filters and any account selection controls display the correct account information.

- Test search, filter behaviour, validation messages, and the empty state when no accounts have been added.

4.4 Categories Module — Approximately 80% Complete

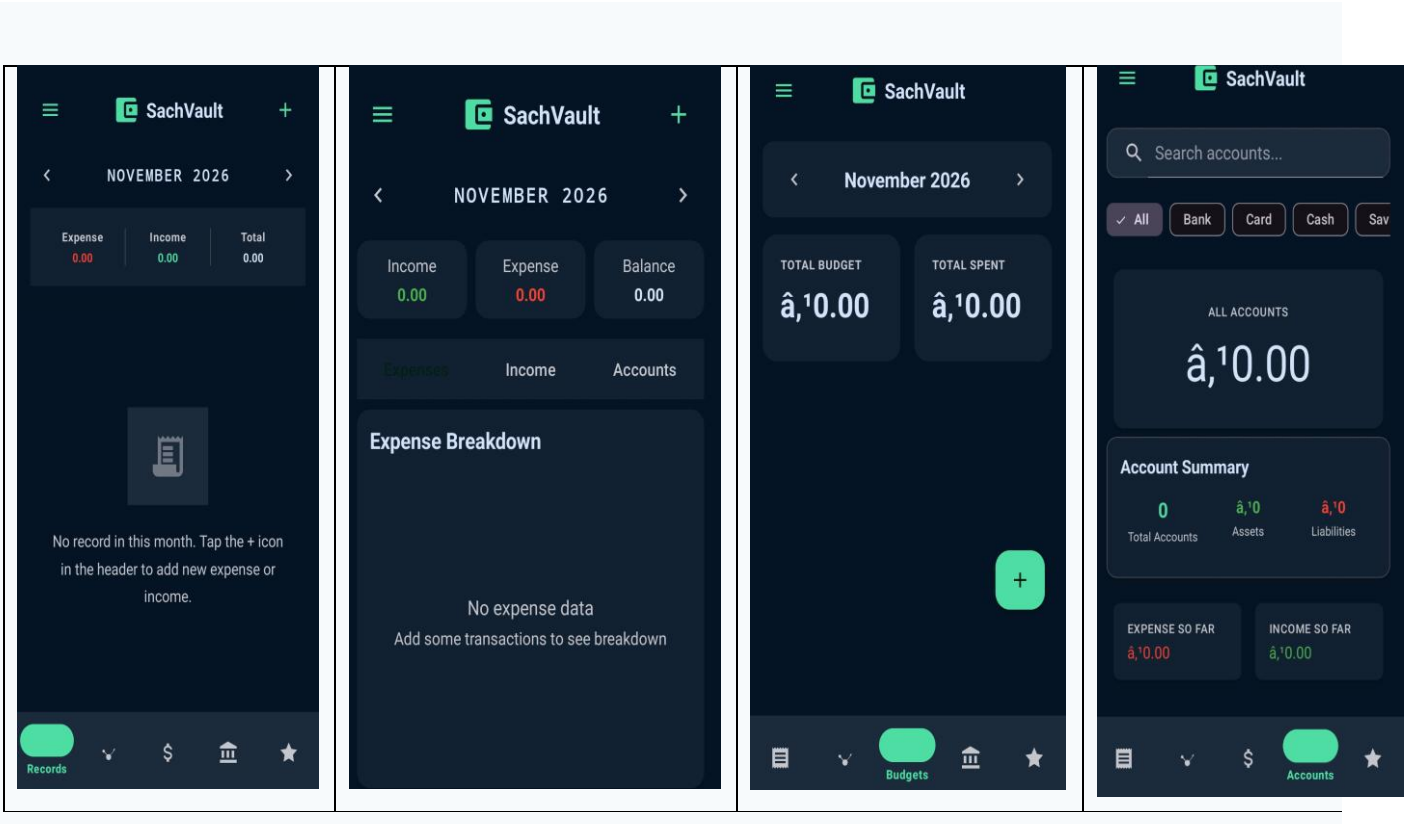
The Categories screen shows an account-balance summary, filters labelled All, Expenses, and Income, and an add button. These controls indicate that the screen is designed to organize categories by transaction type. The module is reported as approximately 80% complete. Remaining work is to complete category creation and management actions and verify that categories are available when recording and reviewing transactions.

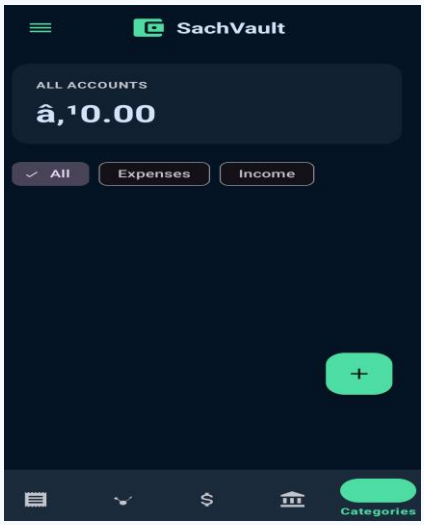
- Complete and verify the category actions opened from the add button.
- Check that the available categories appear correctly in the relevant transaction and category screens.
- Verify that category selection is associated with the correct income or expense records.
- Test validation and empty-state behaviour for category management, including duplicate or blank names if custom categories are supported.

5. Application Screenshots

The following screenshots are from the current application build. They show the Records, Analysis, Budgets, Accounts, and Categories interfaces. The captured screens currently contain mostly zero-value summaries or empty states; this report therefore describes the visible interface and does not assume that sample financial records have already been entered.

Records	Analysis	Budgets	Accounts





6. Technical Implementation

The project is being developed as an Android application using Kotlin for application logic and XML for the screen layouts. The visible screens share a consistent dark interface, bottom navigation, summary cards, filters, and add controls. The Records screen provides a month-based view with income, expense, and total summaries; the Analysis screen presents income, expense, balance, and an expense-breakdown area; the Budgets screen presents total budget and total spent; the Accounts screen provides search and account-type filters; and the Categories screen provides filters for all, expense, and income categories. The screenshots establish the current UI structure. Database technology and persistence behaviour should be described according to the libraries and code actually present in the project.

Typical data flow
User action in Android UI → Kotlin validation and application logic → Room database access (if used) → SQLite local storage → Updated information displayed in the UI.

The implementation description should match the actual source code. Include Room, SQLite, DAO, Repository, or ViewModel details only if those components are configured and used in the current project.

6.1 Data and Calculation Checks

- Enter a small set of test transactions and verify that the Analysis totals and Budget summaries match those records.
- Check that changes made in Records, Budgets, Accounts, or Categories are reflected in the other screens that depend on that information.
- Test amount and other required fields with valid, blank, zero, and invalid input values as appropriate.
- Check that rupee amounts use consistent formatting and that calculations do not introduce unexpected rounding differences.
- Close and reopen the application to verify that entered records and settings are retained by the configured storage layer.

7. Pending Work

Area	Pending tasks
Analysis	Verify Income, Expense, Balance, and Expense Breakdown using entered transactions; check empty-state and refresh behaviour.
Budget	Finish budget add/manage actions and verify Total Budget and Total Spent for the selected month.
Accounts	Finish account-management actions; test search, All/Bank/Card/Cash filters, and data persistence.
Categories	Finish category-management actions; test All/Expenses/Income filters and category use in transactions.
Integration	Check that changes in one module are reflected in dependent screens where applicable.
Testing	Perform functional, validation, persistence, and regression testing.
Documentation	Capture final screenshots and update the report to reflect the tested application.

8. Testing Plan and Verification

The screenshots confirm the presence of the main screens, but they do not by themselves prove that every operation works. Execute each test on the running application and record the result only after checking it.

Test case	Expected result	Result
Open each main tab	Records screen opens and month navigation and summary area display correctly.	Not yet recorded
Budget operation	Budget actions save correctly and Total Budget / Total Spent update as expected.	Not yet recorded
Review analysis	Analysis totals and expense breakdown match the entered records.	Not yet recorded
Account operation	Search and account-type filters show the expected accounts; saved account details persist.	Not yet recorded
Category operation	All, Expenses, and Income filters show the expected categories.	Not yet recorded
Close and reopen app	Previously entered records and settings remain available after restart.	Not yet recorded
Invalid input	Invalid or missing required values are handled with clear validation.	Not yet recorded

9. Next Steps and Completion Plan

1. Complete the remaining work in Accounts and Categories.

2. Complete remaining Budget functionality and verify its calculations.
3. Finish final checks and refinements in Analysis.
4. Test data consistency and interactions across all four modules.
5. Resolve issues found during testing and verify data persistence.
6. Capture final screenshots and update the documentation.
7. Prepare a demonstration covering the implemented workflows.

10. Conclusion

The Phase II screenshots show the current interface for Records, Analysis, Budgets, Accounts, and Categories. The Analysis module is nearly complete, while Budget, Accounts, and Categories are reported at approximately 80% completion. The visible screens provide the structure for reviewing records and financial summaries. The remaining work is to complete the unfinished module operations, verify calculations using entered data, confirm data persistence and cross-screen updates, and carry out final testing.

The project will be considered ready for final demonstration when implemented features work reliably, data is saved and retrieved correctly, and the documentation and screenshots accurately reflect the final application.

Before submitting this report
Replace cover-page placeholders, insert actual screenshots, update test results after running tests, and revise any feature descriptions that do not match the current application. This report intentionally avoids claiming that unverified functions are complete.